

SAN JOSE STATE UNIVERSITY

Degree: Bachelor of Science, Data Science

Timeline: August 2023 - May 2025

Status: GRADUATED (Degree Conferred)

GPA: 3.6/4.0

Relevant Coursework

Course Skills Acquired

Advanced Data Structures and Algorithms Algorithm analysis, trees, graphs, sorting, complexity theory

Processing Big Data (Bash) Command-line data processing, shell scripting, pipeline automation

Probability and Statistics Statistical inference, hypothesis testing, distributions

Advanced Python for Data Analysis Pandas, NumPy, data wrangling, exploratory analysis

Introduction to AI Search algorithms, game theory, basic neural networks

Intro to Design and Analysis of Algorithms Runtime analysis, optimization, divide-and-conquer

Intro to Data Visualization Matplotlib, Seaborn, visual perception principles

Machine Learning Supervised/unsupervised learning, regression, classification, clustering

Database Management (SQL) Query optimization, normalization, relational database design

Advanced Python Object-oriented programming, decorators, generators, APIs

R Programming Statistical computing, ggplot2, dplyr, statistical models

CITY COLLEGE OF SAN FRANCISCO

Degree: Transfer AA in Mathematics

Timeline: January 2012 - May 2023

GPA: 3.88/4.0 (High Honors)

Computer Science Foundation

Course Key Topics

Programming Fundamentals in Java OOP basics, classes, inheritance, polymorphism

Data Structures and Algorithms in Java Linked lists, stacks, queues, trees, hash tables, sorting algorithms

Computer Architecture and Assembly Language MIPS, CPU design, memory hierarchy, registers

Programming Fundamentals in C++ Pointers, memory management, STL, templates

Mathematics Core

Course Applications

Discrete Mathematics Logic, set theory, combinatorics, graph theory foundations

Linear Algebra Matrices, vector spaces, eigenvalues, SVD - essential for ML

Multivariable Calculus Gradients, optimization, backpropagation foundations

Differential Equations Dynamic systems, mathematical modeling

Calculus I and II Limits, derivatives, integrals, series

Physics

Course Concepts

Mechanical Physics Newtonian mechanics, kinematics, forces

Electromagnetic Physics Electricity, magnetism, wave theory

MISSION COLLEGE

Program: CCNA (Cisco Certified Network Administrator)

Year: 2018

Status: Certificate of Completion

Networking Fundamentals

Network protocols and architecture

Routing and switching concepts

Network security basics

TCP/IP stack understanding

Subnetting and IP addressing

Network troubleshooting methodologies

MAJOR PROJECTS DURING ACADEMIC CAREER

GeneScope: MultiModal Deep Learning for Breast Cancer

Timeline: January 2025 - May 2025 (SJSU Capstone)

Links: <https://gene-scope-liard.vercel.app/> | GitHub

Technical Achievement:

Built a full-stack web application for breast cancer stage classification using a MultiModal Deep Neural Network with intermediate fusion. Integrated clinical biomarkers, genomic data, and imaging features.

ML Implementation:

MultiModal DNN architecture combining multiple data types

MLP and CatBoost models optimized for imbalanced datasets

Analytical Hierarchical Process (AHP) for biomarker prioritization

Cox Proportional Hazards regression for prognosis modeling

LLM Integration:

LangChain-based chatbot with Pinecone vector search

Indexed research papers and lecture notes for RAG

Answers gene function and clinical term questions

Skills Applied: PyTorch, Scikit-learn, LangChain, FastAPI, React, Survival Analysis

StairCase: Multiplayer Game with AI Minigames

Timeline: January 2025 - May 2025 (SJSU Project)

Links: <https://stair-case.vercel.app/> | GitHub

Technical Achievement:

Built a cross-platform Snakes & Ladders-style game with real-time multiplayer and AI-powered minigames.

Architecture:

Pygame for game logic and rendering

React frontend on Vercel

Flask backend on NorthFlank via Docker

Firebase Auth + Firestore for user management and real-time state

AI Integration:

LangChain agents with Mistral AI LLM

Dynamic trivia generation

LLM-powered hangman minigame

Skills Applied: React, Flask, Firebase, Docker, LangChain, WebSockets

X-Ray Image Classification

Timeline: SJSU Course Project

Links: GitHub

Technical Achievement:

Trained a hybrid CNN-Bidirectional LSTM network to classify X-ray images for fracture detection and anatomical localization.

Optimization:

Keras Tuner for hyperparameter optimization

Neural Architecture Search (NAS) for optimal structure

Achieved 89.5% test accuracy

Skills Applied: TensorFlow, Keras, CNN, LSTM, Medical imaging

TECHNICAL SKILLS ACQUIRED THROUGH EDUCATION

Programming Languages

Language	Where Learned	Proficiency
Python	SJSU Advanced Python, Self-study	Advanced
Java	CCSF Programming Fundamentals + Data Structures	Advanced
C++	CCSF Programming Fundamentals	Intermediate
JavaScript/TypeScript	Self-study, Projects	Advanced
R	SJSU R Programming	Intermediate
Bash	SJSU Processing Big Data	Intermediate
Assembly (MIPS)	CCSF Computer Architecture	Foundational

ML & Data Science

Skill Source Application

PyTorch	SJSU ML, Self-study	GeneScope deep learning
TensorFlow/Keras	SJSU ML, Projects	X-Ray CNN
Scikit-learn	SJSU ML	Classical ML models
Pandas/NumPy	SJSU Advanced Python	Data analysis
LangChain	Self-study	RAG chatbots
SQL	SJSU Database Management	Data querying

Mathematics Foundation

Area Source ML Application

Linear Algebra	CCSF	Neural networks, embeddings
Calculus	CCSF	Gradient descent, optimization
Probability/Statistics	SJSU, CCSF	Statistical ML, Bayesian methods
Discrete Math	CCSF	Algorithms, graph theory

Networking & Infrastructure

Skill Source Application

Network Protocols	Mission College CCNA	Understanding distributed systems
TCP/IP Stack	Mission College CCNA	API communication
Routing/Switching	Mission College CCNA	Network architecture

ACADEMIC ACHIEVEMENTS

SJSU (2023-2025)

GPA: 3.6/4.0 - Strong performance in upper-division coursework

Completed rigorous Data Science curriculum with emphasis on ML/AI

Capstone project (GeneScope) integrated cutting-edge LLM technology

CCSF (2012-2023)

GPA: 3.88/4.0 - High honors, consistent academic excellence

Completed full Mathematics transfer curriculum

Strong computer science foundation across Java, C++, and assembly

Balanced education while gaining industry experience

Mission College (2018)

CCNA Certificate - Networking fundamentals

Understanding of enterprise network infrastructure

CONNECTION TO CURRENT WORK

How Education Supports Stanford Role

Academic Foundation Applied At Stanford

Advanced Python, Data Structures Building scalable ML pipelines

Machine Learning coursework Understanding ChromBPNet/AlphaGenome internals

Database Management (SQL) BigQuery materialization, data modeling

Processing Big Data (Bash) Data pipeline automation

Linear Algebra/Calculus Gradient-based model interpretation

CCNA Networking Distributed architecture, API communication

LangChain projects LLM agent implementation

React/Flask projects Full-stack platform development

GRADUATION STATUS CONFIRMATION

SJSU B.S. Data Science: COMPLETED (May 2025)

CCSF Transfer AA Mathematics: COMPLETED (May 2023)

Mission College CCNA: COMPLETED (2018)

I am a GRADUATE of all institutions listed. I am not currently enrolled as a student. My academic background provides the theoretical foundation and practical skills I now apply professionally at Stanford School of Medicine.